

ILLIA PIHOVYCH

Backend Engineer · Fractional CTO · AI Product Engineer

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Professional Summary

Backend and product engineer with 16+ years of experience building APIs, automation systems, AI assistants, Telegram bots, data pipelines, admin platforms, and third-party integrations. Strong across JavaScript, TypeScript, Python, PHP, Go, and Rust, with hands-on work in Node.js services, serverless applications, PostgreSQL, MongoDB, Supabase, Linux server administration, cloud deployment, and AI-enabled workflows.

Fractional CTO and independent engineer for products including SalesSlon, ProofDevs, and Znavo. Experienced in turning loosely defined business workflows into reliable backend systems: data models, authentication, webhooks, queues, background workers, scraping pipelines, external API integrations, operational dashboards, and deployment automation.

Backend Strengths

- Designing API-oriented backends and service boundaries for web products, SaaS platforms, bots, and automation tools.
- Building webhook-driven workflows, background workers, queues, scheduled jobs, import pipelines, and retryable operations.
- Modeling data and access control with PostgreSQL, Supabase, MongoDB, MySQL, Firebase, Prisma, and related tools.
- Integrating AI models into production workflows for chat, document analysis, content generation, semantic search, and business automation.
- Connecting products to Telegram, Google services, Stripe, Binance, WooCommerce, WordPress, social platforms, and browser automation pipelines.
- Building admin panels, moderation workflows, role-based portals, audit trails, analytics, and operational tooling around backend systems.
- Deploying and maintaining applications with Vercel, Fly.io, Cloudflare, Docker, GitHub Actions, PM2, Nginx, and Terraform.
- Testing backend behavior with unit, integration, API, and browser-level tests using Vitest and Playwright.

Technical Skills

Languages

JavaScript, TypeScript, Python, PHP, Go, Rust, SQL, PL/pgSQL, C#, Java, PowerShell

Backend and Web

Node.js, Express, Fastify, Next.js server-side applications, REST APIs, WebSockets, serverless functions, Cloudflare Workers, Vercel Functions, background workers, cron jobs, webhook processing, authentication, authorization, and API integrations

Databases and Storage

PostgreSQL, Supabase, MongoDB, Mongoose, MySQL, Firebase, Neon, Prisma, Drizzle ORM, object storage, media pipelines, and import/export workflows

AI and Data Processing

OpenAI API, Google Gemini, Anthropic Claude, Cerebras, Mistral, AI SDK, embeddings, RAG, semantic search, document ingestion, PDF processing, NLP, prompt engineering, AI chatbots, AI content generation, and multi-agent workflows

Automation, Scraping, and Integrations

Puppeteer, Playwright, Cheerio, Crawlee, Chrome DevTools Protocol, stealth browser automation, Google APIs, Google Search and Search Console workflows, Google Sheets and Drive, Telegram Bot API, Telegraf, Grammy, Stripe, Binance API, WooCommerce REST API, Instagram integrations, and WordPress APIs/plugins

Infrastructure and Delivery

Docker, Fly.io, Vercel, Cloudflare, GitHub Actions, PM2, Nginx, SSL, Terraform, environment-based

configuration, CI checks, production troubleshooting, and deployment automation

Systems Administration, DevOps, and CI/CD

Linux server administration, SSH, Nginx, SSL/TLS, DNS, PM2, process supervision, logs, backups, environment configuration, Docker, GitHub Actions, Terraform, automated testing, build pipelines, deployment pipelines, release checks, and production troubleshooting

Cloudflare and PBN Infrastructure

Cloudflare DNS and zone configuration, proxying, SSL/TLS, caching, service setup, deployment of websites, Cloudflare Workers, traffic and domain management, and creation and operation of private blog network (PBN) infrastructure

Testing

Vitest, Playwright, Node.js test runner, API testing, integration testing, end-to-end testing, type checking, linting, import validation, and production smoke checks

Professional Experience

Fractional CTO / Backend & AI Product Engineer

Independent Product Work · 2010–Present

- Own technical direction for software products from architecture and data modeling through backend implementation, integrations, deployment, and operational support.
- Select pragmatic stacks for early-stage products and existing codebases, balancing delivery speed, maintainability, hosting constraints, and future scale.
- Build backend services, admin tools, automation pipelines, AI features, and data workflows that connect business processes across multiple external platforms.
- Work directly with product requirements, translate non-technical goals into implementable systems, and keep delivery focused on useful product behavior.

Selected Fractional CTO Product Work

<https://salesslon.com> - Technical leadership and product engineering for a sales-oriented product family with call recording, AI analysis, management, Telegram bot, and database components. Coordinated backend architecture across separate services for recording workflows, analysis, operational management, and user communication.

<https://proofdevs.top> - Technical direction for a problem-first developer matching and codebase rescue platform. Built the foundation around Next.js, TypeScript, Supabase/PostgreSQL, authentication, row-level security, server-rendered content, admin workflows, employer and freelancer portals, lead handling, moderation, analytics, and immediate content revalidation.

<https://znavo.net> - Fractional CTO product work covering technical architecture, backend planning, product implementation, integrations, and delivery decisions for an independent software product.

Selected Backend Products and Systems

SalesSlon Product Family

Backend product family represented by call recording, AI analyzer, management, Telegram bot, and database components.

- Designed service boundaries for call-related workflows, AI analysis, operational management, and bot-based communication.
- Combined asynchronous processing, external integrations, persistent data, and user-facing administration into a product ecosystem.
- Applied TypeScript and PostgreSQL-oriented development across separate repositories and product surfaces.

Stack: TypeScript, PostgreSQL/PLpgSQL, Node.js, Telegram integrations, AI analysis services

Multi-Bot Telegram Platform

Shared backend and web administration platform for multiple Telegram bots, deployed around Next.js, Vercel, and Supabase.

- Implemented bot registration, lifecycle management, production webhook routing, inbound message persistence, chat resolution, dispatching, queues, retries, and operational state handling.
- Built an administrative inbox with chat previews, bot filtering, pagination, user and operation views, notifications, and moderation workflows.
- Added owner authentication, encrypted webhook handling, media storage, legacy dump import, validation, date repair, and migration tooling.
- Created automated coverage with Vitest and Playwright for backend services, importers, crypto helpers, webhook behavior, admin actions, and end-to-end flows.

Stack: TypeScript, Node.js, Next.js, Supabase, PostgreSQL, Vercel, Tailwind CSS, Vitest, Playwright

ProofDevs Developer Matching Platform

Problem-first platform that turns a plain-language software request into a structured technical interpretation and a shortlist of relevant independent developers.

- Built admin, employer, and freelancer areas with role-aware authentication and protected workflows.
- Implemented developer profiles, skills, verification, publishing, moderation, profile edit approval, leads, chats, blog publishing, analytics, and account management.
- Designed Supabase-backed data access with PostgreSQL, Auth, row-level security, service-side sessions, and server-side mutations.
- Added server-rendered Markdown processing with sanitization, SEO-oriented content generation, ISR, revalidatePath, and dynamic public pages without unnecessary rebuilds.

Stack: TypeScript, Next.js 14 App Router, React, Supabase, PostgreSQL, Tailwind CSS, Zod, WebSockets, server-side Markdown, Vercel

AI Content and SEO Automation Platform

Node.js services for AI-assisted content production, enrichment, reporting, indexing, and WordPress operations.

- Built single-run and worker-based pipelines for site analysis, article generation, multilingual processing, reporting, and content synchronization.
- Integrated Google Gemini, OpenAI, Cerebras, Mistral, Google Trends, WordPress, Supabase, MongoDB, MySQL, and NLP libraries.
- Added background workers, retry handling, language processing, keyword analysis, article cleanup, paid CSV ingestion, and indexing workflows.
- Containerized a production API for Fly.io deployment with Docker and configurable runtime resources.

Stack: Node.js, Express, Gemini, OpenAI, Cerebras, Mistral, MongoDB, Mongoose, MySQL, Supabase, WordPress, Docker, Fly.io

Google Suggestions and Keyword Data System

Large-scale keyword and search suggestion processing system for multilingual SEO research.

- Generated and parsed Google suggestion queries across multiple alphabets, countries, languages, prefixes, and query depths.
- Built restartable and proxy-aware workers with deduplication, queueing, checkpoints, result normalization, and CSV/JSON exports.
- Added browser automation, character encoding handling, translations, MongoDB persistence, statistical reports, and query quality utilities.

Stack: JavaScript, Node.js, Axios, Cheerio, Playwright, Puppeteer, Puppeteer Extra, MongoDB, DeepL, CSV tooling

SEO Audit and Google Indexation Tooling

Command-line and report-generation tools for crawling websites, checking Google indexation, and auditing technical SEO.

- Implemented URL availability checks, Google site: search checks, headless browser checks, sitemap processing, and JSON result persistence.
- Added Crawlee and Cheerio crawlers, configurable CLI execution, audit prompts, AI crawler visibility checks, and HTML report generation.
- Produced reusable workflows for site audits, technical checks, content analysis, and lead-generation research.

Stack: Node.js, Crawlee, Cheerio, Axios, Puppeteer, Google search workflows, JSON, HTML reporting

AI Document and RAG Tools

Local document intelligence systems for ingesting text and PDF files, indexing them, and answering questions with retrieval-augmented generation.

- Implemented document ingestion, PDF extraction, embeddings, vector indexing, semantic retrieval, and command-line querying.
- Connected retrieval workflows to Cerebras and OpenAI-compatible model interfaces while keeping document data local.

Stack: Python, Sentence Transformers, ChromaDB, PDFMiner, OpenAI-compatible APIs, Cerebras, dotenv

Binance and Market Data Services

Backend services and automation tools for cryptocurrency exchange integrations, market metrics, signal workflows, monitoring, and trading-related data processing.

Representative systems include Binance servers, market metrics, taker buy/sell analysis, flux monitoring, signal wizards, price tracking, and exchange-connected browser tools.

Stack: Go, JavaScript, TypeScript, Binance API, HTTP APIs, scheduled workers, market-data processing, dashboards and alerts

Google and Business API Integrations

Built server-side integrations around Google People, Google Sheets, Google Drive, Google Search, Google Search Console, Google Places, and related services.

- Implemented data synchronization, keyword workflows, admin tooling, search and places lookups, spreadsheet-backed operations, and API-oriented internal services.
- Connected external services to bots, SEO systems, content pipelines, and business dashboards.

Stack: JavaScript, TypeScript, Node.js, Google APIs, OAuth-oriented workflows, REST APIs, Sheets, Drive, Search Console, Places

WordPress and PHP Backend Automation

Developed WordPress plugins, PHP extensions, content APIs, migration tools, publishing automation, social posting workflows, and custom site features.

Representative systems include WordPress migration, reviews, author profiles, social posting, homepage management, chat, permalink handling, article export, and GPT-assisted publishing.

Stack: PHP, JavaScript, WordPress APIs, custom plugins, REST APIs, MySQL, Google integrations, OpenAI integrations

Systems Administration, DevOps, CI/CD, and Cloudflare

- Administered Linux servers and production environments with SSH, Nginx, SSL/TLS, DNS, PM2, process supervision, logs, backups, environment configuration, and deployment troubleshooting.
- Designed CI/CD workflows with automated linting, type checking, tests, builds, deployment gates, release checks, and repeatable rollback-ready delivery.
- Used Docker, GitHub Actions, Terraform, Vercel, Fly.io, and Cloudflare to deploy and maintain websites, APIs, workers, and supporting services.
- Configured Cloudflare zones, DNS, proxying, SSL/TLS, caching, traffic routing, and service integrations for production websites.
- Created and operated private blog network (PBN) infrastructure, including website deployment, domain and DNS configuration, Cloudflare setup, content workflows, and monitoring across multiple sites.

Education

BDMA · 1999–2005

Languages

- Ukrainian - Native
- Russian - Native or bilingual proficiency
- English - Professional working proficiency
- Polish - Limited working proficiency